

 poolside/**Laguna-M.1-NVFP4** 

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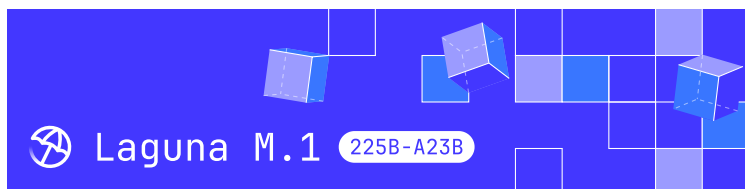
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Laguna M.1-NVFP4

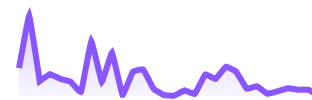
Laguna M.1-NVFP4 is a 225B total parameter Mixture-of-Experts model with 23B activated parameters per token designed for agentic coding and long-horizon work. This is the NVFP4-quantized variant of [Laguna M.1](#).

This is the NVFP4 variant. The [BF16](#) and [FP8](#) variants are also available on Hugging Face.

Highlights

Downloads last month

1,629



 **Safetensors** ⓘ

Model size 131B params

Tensor type F32 · BF16 · F8_E4M3 · U8

 Chat template

↗ Files info

 **Inference Providers** **NEW**

 Text Generation

This model isn't deployed by any Inference Provider.



Ask for provider support

 **Model tree for poolside/Laguna-M...**

Base model poolside/Laguna-M.1

Quantized (13) [this model](#)

 **Collection including poolside/Lag...**

Laguna M.1  Collection

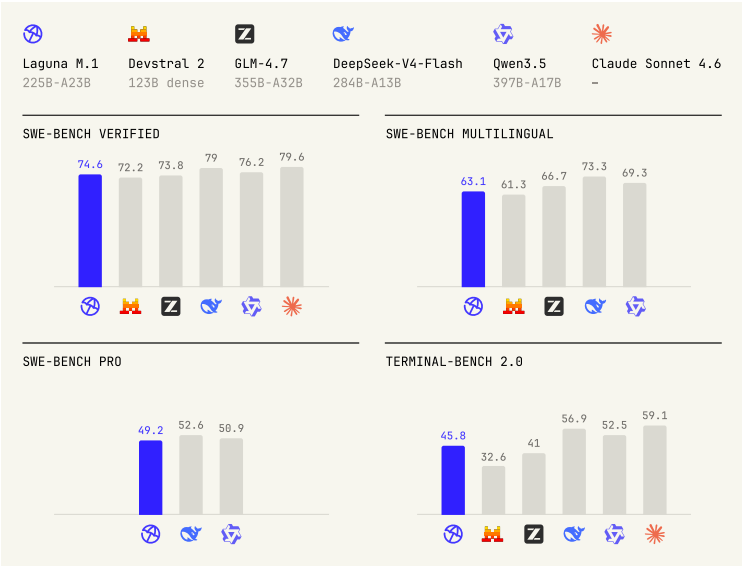
- **Large sparse MoE for agentic coding:** Laguna M.1 is a 70-layer MoE transformer with 225B total parameters and 23B activated parameters per token
 - **High-capacity expert routing:** After 3 dense SwiGLU layers, Laguna M.1 uses 67 sparse MoE layers with 256 experts, top-k=16 routing and auxiliary-loss-free load balancing
 - **Global attention architecture:** Laguna M.1 uses global attention across all layers with 64 Q-heads, 8 KV-heads and softplus attention output gating
 - **Native reasoning support:** Interleaved thinking between tool calls with support for enabling and disabling thinking per-request
 - **Apache 2.0 license:** Use and modify freely for commercial and non-commercial purposes
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Model overview

- Training: pre-training, post-training and reinforcement learning stages
- Number of parameters: 225B total with 23B activated per token
- Optimizer: Muon
- Layers: 70 layers with global attention
- Experts: 256 experts with 1 shared expert; top-k=16 routing
- Dense layers: first 3 layers are dense SwiGLU; remaining 67 layers are sparse MoE

- Attention: 64 Q-heads, 8 KV-heads, head dimension 128, with softplus attention output gating
- Positional encoding: RoPE with YaRN
- Modality: text-to-text
- Context window: 262,144 tokens
- Reasoning support: interleaved thinking with preserved thinking
- Quantization: NVFP4 (weights), detected automatically from quantization_config

Benchmark results



Model	Parameters	SWE-bench Verified	SWE-bench Multilingual	SWE-bench Pro
Laguna M.1 (BF16)	225B-A23B	74.6%	63.1%	49.2%
Devstral 2	123B dense	72.2%	61.3%	-

Model	Parameters	SWE- bench Verified	SWE-bench Multilingual	S b (P Dat
GLM-4.7	355B-A32B	73.8%	66.7%	-
DeepSeek- V4 Flash	284B-A13B	79.0%	73.3%	52.0
Qwen3.5- 397B-A17B	397B-A17B	76.2%	69.3%	50.9
Claude Sonnet 4.6	-	79.6%	-	-

Scores shown are for the BF16 reference model; see the main [Laguna M.1 model card](#) for full benchmarking methodology. We used the highest publicly-referenced scores for all comparison models across each benchmark.

Usage

Laguna M.1 has upstream support in vLLM, SGLang, and TRT-LLM thanks to the support of the team at NVIDIA.

For complete usage instructions, see the main [Laguna M.1 model card](#).

Deployment

vLLM

The full vLLM recipe is on the main [Laguna M.1 model card](#). Quantization is detected automatically

from quantization_config in this checkpoint, so the same command works with poolside/Laguna-M.1-NVFP4 substituted for the model ID. No extra flags required.

```
pip install 'vllm>=0.21.0'
```

```
vllm serve \
  --model poolside/Laguna-M.1-NVFP4 \
  --tool-call-parser poolside_v1 \
  --reasoning-parser poolside_v1 \
  --enable-auto-tool-choice \
  --served-model-name laguna \
  --default-chat-template-kwarg '{"enable_
```

SGLang

The full SGLang recipe is on the [SGLang Cookbook](#).

Quantization is detected automatically, so no extra flags are required.

```
git clone https://github.com/sgl-project/sglang
cd sglang
pip install -e "python[all]"
```

```
sglang serve \
  --model-path poolside/Laguna-M.1-NVFP4 \
  --trust-remote-code \
  --reasoning-parser poolside_v1 \
  --tool-call-parser poolside_v1 \
  --tp 8 \
  --host 0.0.0.0 \
  --port 30000
```

TRT-LLM

Laguna is supported in TensorRT-LLM thanks to the team at NVIDIA ([NVIDIA/TensorRT-LLM#13559](#), with

partial-RoPE fusion in [#15110](#)). The full recipe is on the main [Laguna M.1 model card](#). Quantization is detected automatically from `quantization_config` in this checkpoint, so no extra flags are required.

Controlling reasoning

Laguna M.1 has native reasoning support and is designed to work best with *preserved thinking*, where reasoning content from prior assistant messages is preserved in the message history. This model will generally reason before calling tools and between tool calls. See the main [Laguna M.1 model card](#) for streaming, tool-call, and preserved-thinking examples.

Disabling reasoning

You can disable thinking by setting `enable_thinking` to `False` in a request or by not providing `--default-chat-template-kwarg` `{"enable_thinking": True}` or equivalent when starting the server.

License

This model is licensed under the [Apache 2.0 License](#).

Intended and Responsible Use

Laguna M.1 is designed for software engineering and agentic coding use cases, and you are responsible for confirming that it is appropriate for your intended application. Laguna M.1 is subject to the [Apache 2.0 License](#), and should be used consistently

with Poolside's [Acceptable Use Policy](#). We advise



against circumventing Laguna M.1 safety guardrails without implementing substantially equivalent mitigations appropriate for your use case.

Please report security vulnerabilities or safety concerns to security@poolside.ai.